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Meta- analysis
Dental Implants

Dental implant survival and marginal bone loss after a minimum of 20 years: systematic review and meta-analysis of prospective studies

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Abstract. This systematic review and meta-analysis was performed to assess the 20-year survival rate and marginal bone loss (MBL) of titanium endosseous dental implants with cylindrical or conical screw-shaped designs. Only prospective studies were included. Ten studies met the inclusion criteria. These reported 1857 implants (baseline) of which 1028 were followed up for ≥20 years, during which 115 implant failures were recorded. Implant survival was estimated using a descriptive Kaplan–Meier approach accounting for censored observations. The 20-year implant survival rate was 93.0% (95% confidence interval 91.6–94.1%). The specific cause of failure was explicitly reported for 44 of the 115 implants and was almost equally distributed between biological and mechanical causes. MBL was quantitatively synthesized using a random-effects meta-analysis. The pooled mean MBL was 1.11 mm (95% confidence interval 0.55–1.68 mm). There was substantial variability in long-term marginal bone changes across studies, ranging from slight bone gain to moderate bone loss. Exploratory subgroup analyses suggested differences in MBL according to implant surface characteristics. However, this analysis was based on a limited number of implants and was characterized by high heterogeneity; therefore, the results should be interpreted with caution. The certainty of the evidence (GRADE framework) was rated as moderate for both implant survival and MBL; however, this rating was limited by substantial heterogeneity. Overall, the long-term performance of the dental implants in these prospective studies with follow-up of ≥20 years was favourable. Further well-designed long-term studies are required to elucidate the factors influencing implant survival and marginal bone stability.

Keywords: Dental implants; Survival rate; Epidemiologic measurements; Systematic review; Meta-analysis; Follow-up studies.

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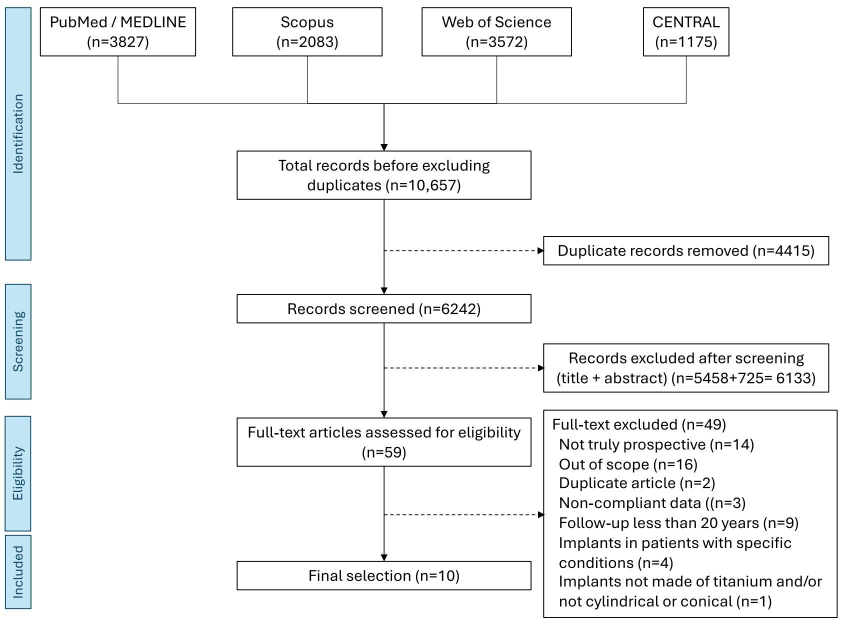

Fig. 1. PRISMA study selection flowchart.

Brånemark began preclinical research on dental implants in the 1960s and published the first human data in 1977; since then, dental implants have been used successfully for over half a century[1,2]. Despite their high success rates[3], dental implants still present certain weaknesses that can lead to failure of the implant-supported prosthetic rehabilitation. Among the causes of failure are complications related to the prosthetic or implant components, as well as the loss of osseointegration due to inflammation of the peri-implant tissues. The literature reports particularly high incidence rates for such complications, with up to 48% of implants affected by peri-implant mucositis and 6.6–12.4% affected by peri-implantitis[4–6]. These data have generated substantial uncertainty, particularly among clinicians, triggering wide-ranging debate[7–19]. The initial concern has been partially mitigated by the contributions of several authors who have critically interpreted the data and clarified that most of the failures observed are attributable to poor-quality dentistry—both in terms of clinical execution and materials used[20]. Another confounding factor in accurately assessing the effectiveness of implant-supported rehabilitations is the abundance of low-quality studies in the literature, with relatively few investigations meeting high scientific standards[21–23]. Moreover, most available studies have reported short- or very short-term outcomes, with follow-up of less than 5 years, while both patients and clinicians are increasingly interested in long- and very long-term follow-up data. This is particularly relevant given the increasing life expectancy, which has led implant patients to demand extended longevity from their prosthetic solutions[24]. Thus, the question of the true long-term effectiveness of implant-supported prosthetic rehabilitations remains open.

This leads to the general question, "What are the long-term outcomes of dental implant treatment?" The aim of this systematic review and meta-analysis was to answer the following more specific question: "In adults receiving screw-shaped titanium endosseous dental implants, what are the 20-year survival rate and marginal bone changes as reported in prospective studies and randomized clinical trials?".

Materials and methods

This systematic review was conducted in accordance with the updated 2020 PRISMA statement[25]. The study protocol was registered prospectively in the PROSPERO database (CRD42024597694).

Eligibility criteria

Since this study was not searching for studies that compared different interventions, the following PIO criteria were used to evaluate the relevance of the studies in relation to the research question: the population (P) comprised adult patients (≥18 years old), representative of the general population undergoing implant treatment; the intervention (I) was the placement of commercially pure titanium or grade V titanium implants, with a cylindrical or conical screw-shaped design, in native maxillary or mandibular bone; the primary outcome (O) was the implant survival rate at 20 years, with the implant as the unit of analysis. The secondary outcome was marginal bone loss (MBL) at 20 years, also measured per implant.

The exclusion criteria were studies involving patients with specific conditions (e.g., osteoporosis, prior radiotherapy); studies involving zirconia implants, implant surfaces not based on TiO_2, and zygomatic, pterygoid, blade, or needle implants; studies not reporting the primary outcome with a 20-year follow-up; non-prospective studies.

Information sources, search strategy, and selection process

The following electronic databases were searched: PubMed (MEDLINE), Scopus, Web of Science, and Cochrane CENTRAL. The last search was conducted on December 10, 2024.

The search combined both free-text and medical subject heading (MeSH) terms. The full search strategy is provided in Supplementary material Table S1.

Zotero (version 6.0.37; https://www.zotero.org/) was used for reference management. After duplicate removal (J.A.), two reviewers (G.I., S.M.L.) independently screened the titles and abstracts for relevance. Full-text assessment was performed for studies meeting the inclusion criteria. A full-text review was also conducted to reach a decision regarding inclusion if the title and abstract did not provide sufficient information to determine eligibility. Disagreements were resolved by discussion. The full texts of the remaining studies were assessed based on the inclusion criteria defined by the PIO framework. The study selection flow diagram is shown in Fig. 1.

Data collection

The primary and secondary outcomes were extracted independently by two reviewers (J.A., S.M.L.) using Microsoft Excel (v.16.80; Microsoft Corp., Redmond, WA, USA). Discrepancies were resolved through discussion.

The primary outcome considered in this review was the implant survival rate, calculated as the number of surviving implants relative to the number of implants placed, and reported both as a percentage and as absolute values. The implant was the statistical unit. The numbers of implants at risk, failures, and censored observations were extracted for each yearly interval from the original studies. Failure was defined as implant removal for any reason. Implants lost to follow-up, patients who died, and implants not evaluated at a given time point were treated as censored observations at the time of last observation.

The secondary outcome was MBL, expressed in millimetres (mean ± standard deviation), recorded at the main time point of 20 years and at any intermediate time points if available. MBL was defined as the change in marginal bone level over time relative to the baseline examination; positive values indicate bone loss, while negative values indicate marginal bone gain.

Additional data collected included the study year, setting, and country; implant surface treatment and design; the items used for assessing methodological quality (MINORS); number of censored implants; number of patients at baseline (T0) and at 20 years (T20); number of dropouts; and causes of implant failure and the number of patients who experienced implant failure.

In the case of missing or inconsistent data, the corresponding author of the study in question was contacted via email.

Risk of bias assessment

As the included studies were not necessarily randomized, the MINORS index was applied to assess the risk of bias[26]. The evaluation included the following criteria: a clearly stated aim, consecutive patient inclusion, prospective data collection, clearly defined endpoints, unbiased endpoint assessment, an appropriate follow-up period relative to the aim of the study, and adequate loss to follow-up. For the MINORS assessment, the item ‘prospective calculation of the study size’ was operationalized as sample size adequacy, as a prospective sample size calculation was not applicable to the long-term observational cohort studies included. Assuming an expected implant survival of 95%, a sample of approximately 75 implants would allow estimation with a 95% confidence interval of ± 5%; therefore, studies were scored as adequate if they included > 75 implants (score = 2), suboptimal if they included 50–75 implants (score = 1), and insufficient if they included < 50 implants (score = 0). Based on the total MINORS score for non-comparative studies (maximum score = 16), the included studies were classified as low risk of bias (≥13), moderate risk of bias (9–12), or high risk of bias (≤8).

Effect measures

For MBL, the estimated mean derived from a meta-analysis was used as the effect measure. In the case of stratified analyses, the synthesis was based on comparisons of estimated survival rates across subgroups.

Synthesis methods

Eligibility for synthesis was determined by tabulating study characteristics and comparing them to planned synthesis groups. Data were prepared for presentation without modifications beyond those required for synthesis. When possible, missing data were inferred from the information provided in the included studies. Implant survival over the 20-year follow-up period was estimated using the Kaplan–Meier method, with implants lost to follow-up treated as censored observations at the time of last observation. Survival probabilities were calculated annually using the product-limit estimator. Ninety-five percent confidence intervals (95% CI) were computed using Greenwood’s formula. The survival curve was generated using aggregated time-to-event data reconstructed from the included studies. A meta-analysis was conducted for the synthesis of results when the available data allowed. A random-effects model was applied (DerSimonian and Laird method), as the inclusion criteria allowed for the incorporation of studies from highly heterogeneous clinical and methodological settings.

The data analysis was conducted using R Studio (version 2024.12.1+563; Posit Software). Statistical heterogeneity was evaluated using I^2 and τ^2. Subgroup analyses were performed to explore sources of heterogeneity.

Sensitivity analyses, reporting bias, and certainty of the evidence

Sensitivity analysis was conducted by excluding studies deemed at high risk of bias based on the MINORS criteria.

Reporting bias was assessed for pooled outcomes using funnel plots and Egger’s test. The assessment of reporting bias was not applied to survival outcomes estimated using the Kaplan–Meier method.

The certainty of the evidence was evaluated using the GRADE approach[27].

Results

Study selection

The literature search retrieved 3827 records from PubMed/MEDLINE, 2083 records from Scopus, 3572 records from Web of Science, and 1175 records from Cochrane CENTRAL. No additional studies were identified through manual reference searching. A total of 10,657 records were initially identified prior to duplicate removal. After removing 4415 duplicate entries, 6242 unique records remained for screening. Following title screening, 5458 records were excluded, leaving 784 records for abstract screening. Of these, 725 were excluded based on abstract evaluation, resulting in 59 full-text articles assessed for eligibility. Forty-nine of these full-text articles were excluded due to not meeting the eligibility criteria. Consequently, 10 studies were included in this systematic review and meta-analysis[28–37].

The PRISMA flow diagram detailing the study selection process is presented in Fig. 1.

Study characteristics

The included studies were published between 2003 and 2024 and conducted across seven different countries: Sweden (four studies), Italy, Belgium, Germany, the Netherlands, Switzerland, and Canada. The implants featured either a microrough surface (TiUnite, Nobel Biocare; SLActive, Straumann; TiOblast, Astra Tech; TPS, Dentsply Friadent; TPS, Institut Straumann) or machined surface (machined surface, Nobel Biocare; machined surface, Astra Tech). Three studies assessed microrough surfaces[31,36,37], four studies evaluated machined surfaces[28–30,34], and three studies included both surface types[32,33,35]. Two studies were conducted in private practice settings[36,37], while eight were conducted in university hospitals[28–35]. The characteristics of the included studies are detailed in Table 1.

Table 1. Characteristics of the included studies.

Study	Surface	Implant	Setting	Country
Östman et al.[37] 2024	Microporous anodized (TiUnite)	Brånemark (Nobel Biocare) MkIII and MkIV	Private practice	Sweden
Roccuzzo et al.[36] 2022	Sandblasted large grit and acid-etched surface (SLActive)	Straumann Group AG	Private practice	Italy
Jacobs et al.[35] 2021	TiO2-blasted; machined	Astra Tech; Brånemark (Nobel Biocare) MkII	University hospital	Belgium
Thöne-Mühling et al.[34] 2020	Machined	Brånemark (Nobel Biocare) MkII	University hospital	Germany
Bakker et al.[33] 2019	Machined; titanium plasma-sprayed (TPS)	Brånemark (Nobel Biocare); IMZ (Dentsply Friadent); ITI (Institut Straumann);	University hospital	Netherlands
Donati et al.[32] 2018	Machined; TiO2-blasted	Astra Tech	University hospital	Sweden
Chappuis et al.[31] 2013	Titanium plasma-sprayed (TPS)	Bonefit System (Institute Straumann)	University hospital	Switzerland
Åstrand et al.[30] 2008	Machined	Brånemark (Nobel Biocare)	University hospital	Sweden
Attard and Zarb[29] 2004	Machined	Brånemark (Nobel Biocare)	University hospital	Canada
Ekelund et al.[28] 2003	Machined	Brånemark (Nobel Biocare)	University hospital	Sweden

Risk of bias in studies

Eight of the included studies showed a moderate risk of bias according to the MINORS criteria, indicating medium methodological quality. Two studies presented a low risk of bias, reflecting high methodological quality. No study was rated as having a high risk of bias. A summary of the risk of bias analysis is given in Fig. 2.

Results of individual studies

The 10 included studies reported a total of 1857 implants placed in 614 patients at baseline (T0). At 20 years of follow-up (T20), 1028 implants and 342 patients

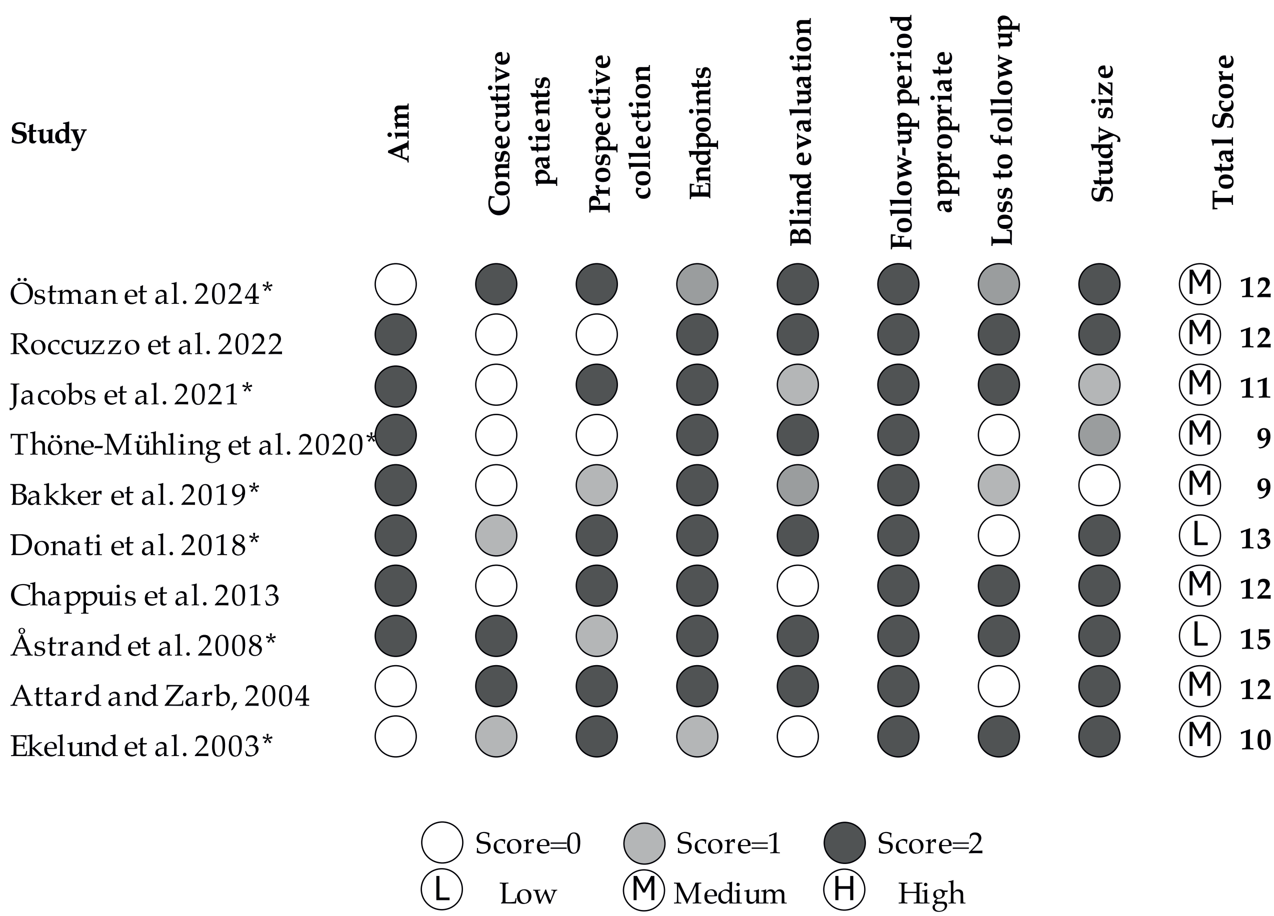

Fig. 2. Risk of bias of the included studies according to the MINORS Index. Each item was scored from 0 (not reported) to 2 (adequately reported). Based on the total score, studies were classified as low (≥13), medium (9–12), or high (≤8) risk of bias. Studies marked with an asterisk were included in the meta-analysis of MBL.

Table 2. Results of the individual studies.

Study	Implants, *n*		Censures	Failures	MBL		Patients, *n*	
	T0	T20			Mean	SD	T0	T20
Östman et al.[37]	133	121	5	7	−0.54	1.19	66	39
Roccuzzo et al.[36]	297	160	125	12	NA	NA	149	84
Jacobs et al [35].	95	51	44	0	0.13[a]; 0.35[b]	0.57[a]; 0.78[b]	18	10
Thöne-Mühling et al.[34]	126	56	61	9	1.98	1.64	39	19
Bakker et al.[33]	106	22	76	8	1.14	0.85	53	15
Donati et al.[32]	148	64	66	18	0.83[a]; 0.41[b]	1.59[a]; 1.25[b]	51	25
Chappuis et al.[31]	145	85	50	10	NA	NA	98	67
Åstrand et al.[30]	269	123	132	14	−2.33	0.13	48	21
Attard and Zarb[29]	265	167	64	34	NA	NA	45	32
Ekelund et al.[28]	273	179	91	3	1.6	0.9	47	30
Total	1857	1028	714	115			614	342

MBL, marginal bone loss; NA, data not available; SD, standard deviation; T0, baseline; T20, 20-year follow-up.
[a]Microrough implants.
[b]Machined implants.

remained; there were 115 implant failures and 714 censored implants. A summary of the results of the individual studies, including the numbers of implants and patients in each, is provided in Table 2.

Survival rate

The Kaplan–Meier survival analysis accounting for censored observations yielded a 20-year implant survival rate of 93.0% (95% CI 91.6–94.1%) (Table 3; Fig. 3).

Among the 115 implant failures recorded across the included studies, the specific cause of failure was explicitly reported for 44. These failures were almost equally distributed between biological causes (including peri-implantitis and loss or lack of osseointegration) and mechanical causes (primarily implant fracture). The original studies did not provide any information for the other failures to allow a reliable classification.

Marginal bone loss (MBL)

MBL was quantitatively synthesized using a random-effects model on measurements available at the end of observation, including only implants with at least 20 years of follow-up. Seven studies contributed to the pooled analysis, with MBL expressed as an aggregated mean in millimetres. In three studies, data were reported for distinct experimental subgroups and these were treated as separate datasets[32,34,35]. The pooled mean MBL across all included studies was 1.11 mm (95% CI 0.55–1.68 mm) at 20 years; the corresponding 95% prediction interval ranged from −0.49 mm to 2.72 mm, indicating substantial variability in long-term bone changes across studies. The heterogeneity observed among studies was considerable (I^2 = 99.1%) (Fig. 4). Subgroup analysis according to implant surface (microrough vs machined) revealed a statistically significant difference in pooled MBL values (P = 0.015), with higher mean MBL observed for machined surfaces (1.43 mm, 95% CI 0.52 mm to 2.33 mm) compared with microrough surfaces (0.45 mm, 95% CI −0.39 mm to 1.29 mm) (Fig. 5). However, high heterogeneity persisted within both subgroups, hence these findings were considered exploratory. Subgroup analysis based on study quality, assessed using the MINORS score, did not demonstrate statistically significant differences in MBL between studies classified as low or medium risk of bias (P = 0.82). Similarly, subgroup comparisons according to study setting were not considered informative, as only one study was conducted in a private practice setting, precluding meaningful comparisons.

Table 3. Survival rates as per Kaplan–Meier analysis.

Year	Implants at risk	Failures	Censored	Survival (%)	95% CI	
					Lower	Upper
1	1857	0	0	100	100	100
2	1815	30	12	98.35	97.64	98.84
3	1806	6	3	98.02	97.27	98.57
4	1802	4	0	97.8	97.02	98.38
5	1766	16	20	96.92	96.01	97.62
6	1763	0	3	96.92	96.01	97.62
7	1760	1	2	96.86	95.95	97.57
8	1743	6	11	96.53	95.58	97.28
9	1740	3	0	96.36	95.39	97.13
10	1635	15	90	95.48	94.41	96.35
11	1595	0	40	95.48	94.41	96.35
12	1539	9	47	94.92	93.79	95.85
13	1528	1	10	94.86	93.72	95.79
14	1524	2	2	94.73	93.58	95.68
15	1485	2	37	94.61	93.44	95.57
16	1478	2	5	94.48	93.3	95.45
17	1472	0	6	94.48	93.3	95.45
18	1463	4	5	94.22	93.01	95.22
19	1458	1	4	94.15	92.94	95.16
20	1028	13	417	92.96	91.58	94.13

CI, confidence interval.

Reporting bias

The funnel plot for MBL showed no evidence of marked asymmetry (Supplementary material Fig. S1). Egger's test did not detect statistically significant funnel plot asymmetry for MBL (z = 1.49, P = 0.14), although the limited number of studies should be considered when interpreting this result.

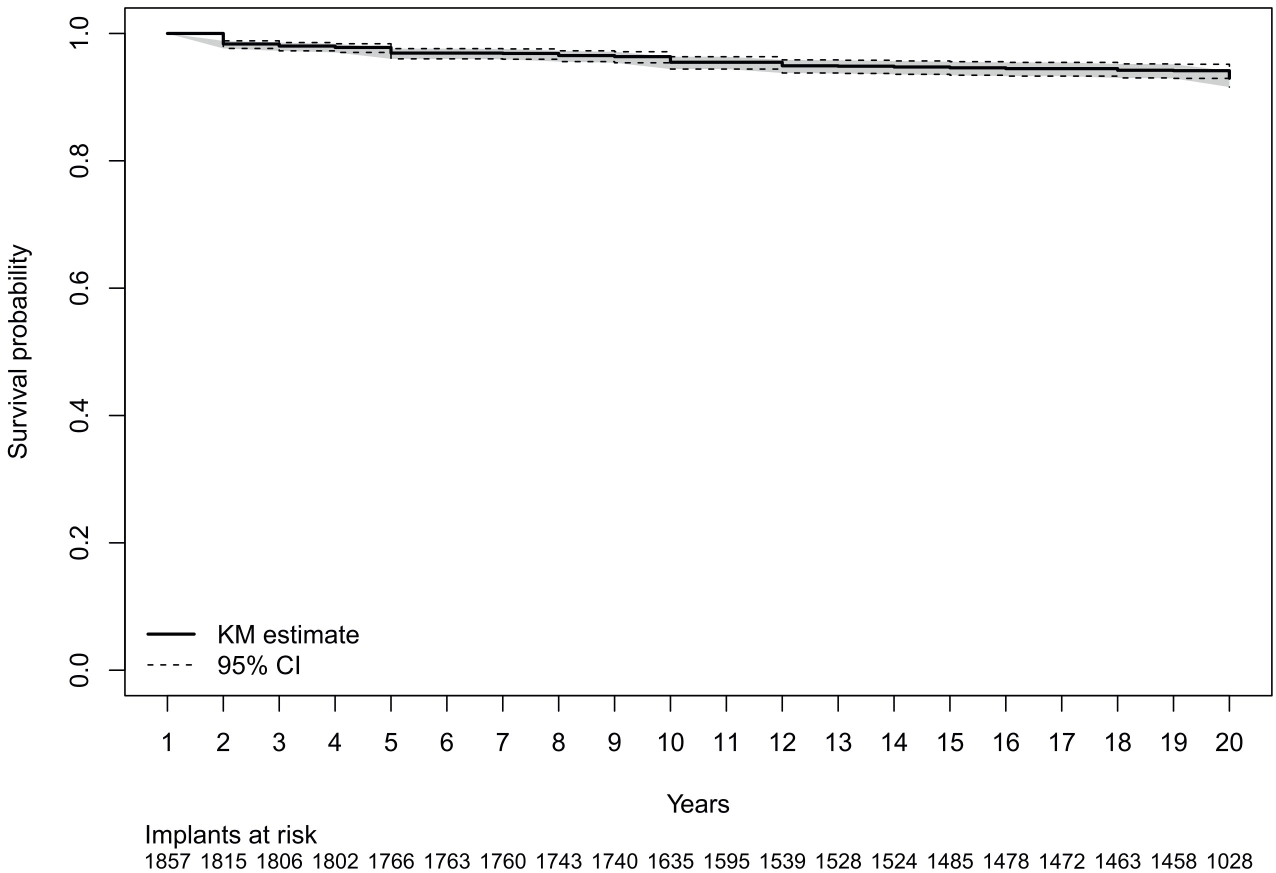

Fig. 3. Kaplan–Meier survival curve for dental implant survival over 20 years. The shaded areas represent the 95% confidence interval. The number of implants at risk at each time point is shown below the *x*-axis.

Certainty of the evidence

Certainty in the body of evidence was assessed according to the GRADE framework[27]. For this assessment, risk of bias was evaluated using the MINORS tool[26], inconsistency through heterogeneity analysis, indirectness by assessing differences in study populations, interventions, and outcome definitions, imprecision by the width of the confidence intervals, and reporting bias via funnel plot inspection and Egger's regression test. The initial level of the quality of evidence was considered high because the included non-randomized studies were adequately assessed for risk of bias using a tool that assesses risk of bias along an absolute scale[38]. All outcomes showed substantial heterogeneity. Indirectness was judged as not serious, as differences

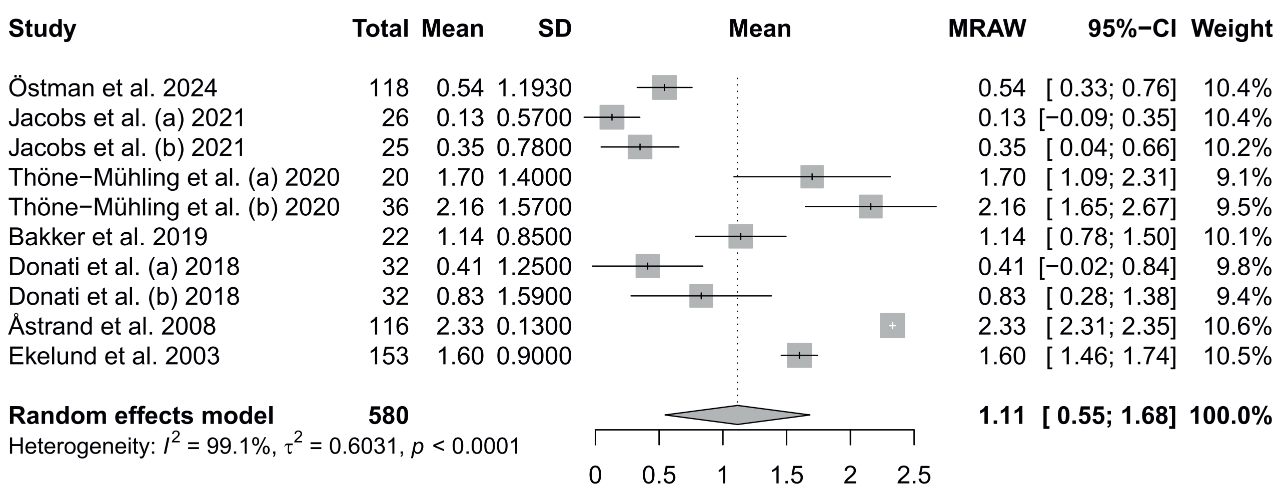

Fig. 4. Random-effects meta-analysis of marginal bone loss (MBL) at 20 years. The squares represent the individual study estimates with 95% confidence intervals. The diamond indicates the pooled mean estimate. In three studies, the outcome was reported separately for two subgroups; these were extracted and analysed as independent data points, as presented in the original publications.

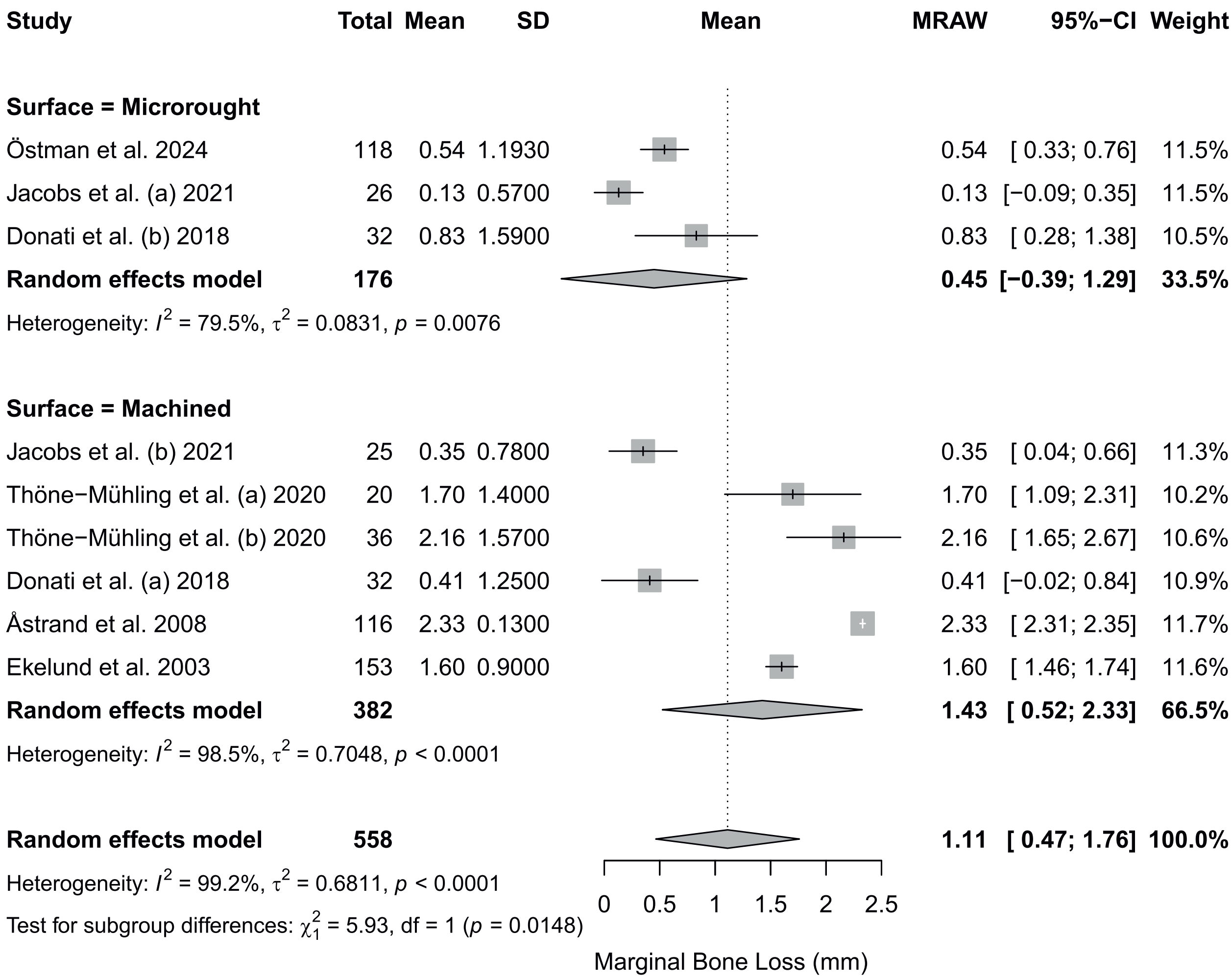

Fig. 5. Forest plot of marginal bone loss (MBL) stratified by implant surface.

among studies in terms of populations, interventions, and outcome definitions were not considered significant. Imprecision was judged as not serious, as confidence intervals were considered narrow for both implant survival and MBL, taking into account the magnitude of the estimates and their contextualization within the 20-year follow-up period. The overall certainty of the evidence, assessed using the GRADE framework, was rated as moderate for both implant survival and MBL (Table 4).

Discussion

The aim of this study was to update current evidence by conducting a systematic review and meta-analysis to determine implant survival rates at 20 years, as well as the long-term marginal bone loss (MBL) associated with implant function. This study provides an up-to-date comprehensive synthesis of all prospective data available on 20-year implant survival. Ten rigorous prospective studies were included, accounting for 1857 implants and 614 patients at baseline, with 1028 implants and 342 patients remaining at the end of the follow-up period. These data reflect both the high methodological quality of the included studies and the inherent challenges in conducting prospective studies over such an extended duration. Using a Kaplan–Meier approach, the estimated implant survival was 93.0% (95% CI 91.6–94.1%), while the overall MBL was 1.11 mm (95% CI 0.55–1.68 mm).

Table 4. GRADE—certainty of the evidence.

Outcome	Study design	Risk of bias (MINORS)	Inconsistency	Indirectness	Imprecision	Publication bias	Overall certainty of the evidence
Implant survival at 20 years[a]	RCTs and NRS	Moderate to low[1]	Serious	Not serious	Not serious	Not detected	Moderate
MBL at ≥20 years (mm)	RCTs and NRS	Moderate to low[1]	Serious	Not serious	Not serious	Not detected	Moderate

MBL, marginal bone loss; NRS, non-randomized study; RCT, randomized clinical trial. Superscript ‘1’ indicates that the evidence was downgraded by one level.
[a]Kaplan–Meier estimate.

Studies in the literature have reported implant survival rates exceeding 96% at 5 years and 89% at 10 years[39–41]. Compared to these, the survival rate observed at 20 years in the current meta-analysis remains within the range of previously reported long-term outcomes. The present findings align with those of a previous meta-analysis that reported a 10-year survival rate of 93.2% from prospective studies[42]. Although the overall implant survival rate appears high, it indicates that long-term survival is not universal, with a small but clinically relevant proportion of implants not remaining in function even in well-selected patient cohorts. This finding challenges overly optimistic claims of 100% long-term implant survival. To simplify, assuming a constant failure rate over time, a 20-year survival rate of 93% corresponds to a crude average annual failure rate of approximately 0.35%, equivalent to about one implant failure per 286 implants per year.

Differentiation between biological and technical causes of implant failure is of considerable clinical and industrial relevance. However, in the present review, detailed information on the cause of failure was available for less than half of the failed implants. Among the reported cases, biological and mechanical failures were almost equally represented, suggesting that long-term implant failure may result from both biological complications and mechanical limitations. In other terms, if it were possible to eliminate failures due to implant fracture, the failure rate would be reduced by half.

Given the substantial heterogeneity observed across the MBL outcome, stratified analyses were conducted considering three variables for MBL: implant surface type, study quality, and clinical setting. Implant surface characteristics are generally considered an important factor influencing long-term outcomes, but their exact role has not been fully established. The first implants introduced by Brånemark featured machined surfaces[43,44]; later, microrough surfaces were developed to enhance and accelerate osseointegration. More recently, novel surface technologies have been introduced, some of which challenge the assumption that increased roughness is inherently beneficial[45].

The stratified analysis by implant surface type suggested reduced long-term MBL for microrough surfaces. Although machined implants showed slightly less favourable MBL values, these remained within the clinically acceptable range. However, the estimates were based on a relatively small pooled sample of 558 implants, and high heterogeneity persisted within both subgroups; therefore, the findings should be interpreted as exploratory. Accordingly, the results of this meta-analysis do not provide strong support for the preferential adoption of microrough surfaces, and implant surface selection should instead be guided by broader clinical considerations[46].

The stratified analysis according to study quality did not identify significant differences, and the analysis by study setting was not considered informative. These findings are based on a limited number of studies and do not allow definitive conclusions, underscoring the need for further studies specifically addressing these factors.

Rather than supporting inferential conclusions, these observations should be considered hypothesis-generating, indicating that under optimal and well-controlled conditions peri-implant bone levels may be maintained or even improved, in contrast to criteria that typically assume some degree of bone loss over time[47]. The mean MBL of 1.43 mm at 20 years observed for machined implants is also remarkably favourable when compared to thresholds commonly accepted in the literature. Based on Albrektsson's criteria, implants remain successful with bone loss below 5.3 mm at 20 years; both the bone gain in high-quality studies and the limited resorption in medium-quality ones fall well within this limit[47]. This suggests that, in ideal clinical scenarios, peri-implant bone resorption may be extremely limited even in the very long term[20].

A recently published meta-analysis reported a 92% survival rate based on three prospective studies and 237 implants[48]. In the present meta-analysis, the corresponding survival rate was very similar (93.0%). However, it is important to note that this analysis included a broader dataset of 10 studies and 1143 implants—approximately quintupling the sample size of the prior work. Furthermore, the previous meta-analysis did not assess MBL.

From a methodological standpoint, only strictly prospective studies were included, all demonstrating high methodological rigor. According to the MINORS assessment, none of the included studies was classified as high risk for bias. The fact that only 10 studies met the inclusion criteria out of over 10,000 initially screened suggests that high-quality long-term studies remain scarce, highlighting the need for more robust long-term clinical research in implant dentistry.

The high degree of heterogeneity for the MBL outcome—unresolved even after stratified analysis—may be attributed to unmeasured or unreported risk factors in the included studies. Although this meta-analysis focused on healthy patients, stratification by other recognized risk factors (e.g., smoking, diabetes) or demographic variables (e.g., age, sex) was not feasible due to data limitations. Given the extended follow-up, it is also plausible that initially healthy patients developed systemic conditions over time (e.g., cancer), which may have negatively impacted implant survival[49], further reducing the survival rate.

In this study, implant survival was the primary outcome, while the incidence rates of mucositis and peri-implantitis were not directly evaluated. MBL was therefore considered as an indirect marker of chronic peri-implant inflammation. Notably, the MBL values observed in this meta-analysis were low and remained well below physiological thresholds. Several clinically relevant variables, including patient characteristics, implant and prosthetic designs, and treatment protocols, were inconsistently reported across studies and could therefore not be included in subgroup or meta-regression analyses, limiting the ability to fully explain between-study heterogeneity.

Two of the included studies were published more than 20 years ago, which may represent a potential limitation. On the other hand, these studies fully met the predefined inclusion criteria, and the implant survival rate is not a particularly complex outcome nor one for which substantial methodological changes have occurred since the time of publication. Moreover, the older studies exclusively evaluated implants with machined surfaces, and a sensitivity analysis addressing this specific factor was performed.

Improved reporting of failure mechanisms in future long-term prospective studies would be particularly valuable for clinicians and manufacturers aiming to optimize implant designs and long-term performance. Future studies with more standardized selection criteria and better control of methodological variables may improve estimate precision and reduce observed heterogeneity.

Ethical approval

Not required.

Patient consent

Not required.

Review registration

This review is registered in PROSPERO (CRD42024597694; https://www.crd.york.ac.uk/prospero/display_record.php?ID=CRD42024597694).

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None.

Competing interests

None.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.ijom.2026.05.011.

References

1. Brånemark PI, Hansson BO, Adell R, Breine U, Lindström J, Hallén O, et al. Osseointegrated implants in the treatment of the edentulous jaw. Experience from a 10-year period. *Scand J Plast Reconstr Surg Suppl* 1977;**16**:1–132.
2. Brånemark PI, Adell R, Breine U, Hansson BO, Lindström J, Ohlsson A. Intra-osseous anchorage of dental prostheses. I. Experimental studies. *Scand J Plast Reconstr Surg* 1969;**3**(2):81–100. https://doi.org/10.3109/02844316909036699
3. Rizzo S, Zampetti P, Baena RRY, Svanosio D, Lupi SM. Retrospective analysis of 521 endosseous implants placed under antibiotic prophylaxis and review of literature. *Minerva Stomatol* 2010;**59**(3):75–88.
4. Roos-Jansåker A-M, Lindahl C, Renvert H, Renvert S. Nine- to fourteen-year follow-up of implant treatment. Part II: presence of peri-implant lesions. *J Clin Periodontol* 2006;**33**(4):290–5. https://doi.org/10.1111/j.1600-051X.2006.00906.x
5. Fransson C, Lekholm U, Jemt T, Berglundh T. Prevalence of subjects with progressive bone loss at implants. *Clin Oral Implants Res* 2005;**16**(4):440–6. https://doi.org/10.1111/j.1600-0501.2005.01137.x
6. Koldsland OC, Scheie AA, Aass AM. Prevalence of peri-implantitis related to severity of the disease with different degrees of bone loss. *J Periodontol* 2010;**81**(2):231–8. https://doi.org/10.1902/jop.2009.090269
7. Schmalz G, Ziebolz D. The hypotheses of oral disease-associated perioprosthetic joint infections to understand the role of oral health for endoprostheses—a narrative review. *Spec Care Dentist* 2024;**44**(2):328–38. https://doi.org/10.1111/scd.12901
8. Ardila CM, Vivares-Builes AM. Antibiotic resistance in patients with peri-implantitis: a systematic scoping review. *Int J Environ Res Public Health* 2022;**19**(23):15609. https://doi.org/10.3390/ijerph192315609
9. Monje A, Amerio E, Farina R, Nart J, Ramanauskaite A, Renvert S, et al. Significance of probing for monitoring peri-implant diseases. *Int J Oral Implantol Berl Ger* 2021;**14**(4):385–99.
10. Dawoud BES, Kent S, Tabbenor O, George P, Dhanda J. Dental implants and risk of bleeding in patients on oral anticoagulants: a systematic review and meta-analysis. *Int J Implant Dent* 2021;**7**(1):82. https://doi.org/10.1186/s40729-021-00364-5
11. Al-Sabbagh M, Shaddox LM. Is peri-implantitis curable? *Dent Clin N Am* 2019;**63**(3):547–66. https://doi.org/10.1016/j.cden.2019.02.003
12. Aghaloo TL, Mardirosian M, Delgado B. Controversies in implant surgery. *Oral Maxillofac Surg Clin N Am* 2017;**29**(4):525–35. https://doi.org/10.1016/j.coms.2017.07.007
13. Ata-Ali J, Ata-Ali F, Bagan L. A classification proposal for peri-implant mucositis and peri-implantitis: a critical update. *Open Dent J* 2015;**9**:393–5. https://doi.org/10.2174/1874210601509010393
14. Seki K, Nakabayashi S, Tanabe N, Kamimoto A, Hagiwara Y. Correlations between clinical parameters in implant maintenance patients: analysis among healthy and history-of-periodontitis groups. *Int J Implant Dent* 2017;**3**(1):45. https://doi.org/10.1186/s40729-017-0108-0
15. Valente NA, Andreana S. Peri-implant disease: what we know and what we need to know. *J Periodontal Implant Sci* 2016;**46**(3):136–51. https://doi.org/10.5051/jpis.2016.46.3.136
16. Monje A, Insua A, Wang HL. Understanding peri-implantitis as a plaque-associated and site-specific entity: on the local predisposing factors. *J Clin Med* 2019;**8**(2):279. https://doi.org/10.3390/jcm8020279
17. Coli P, Sennerby L. Is peri-implant probing causing over-diagnosis and over-treatment of dental implants? *J Clin Med* 2019;**8**(8):1123. https://doi.org/10.3390/jcm8081123
18. Fragkioudakis I, Tseleki G, Doufexi A-E, Sakellari D. Current concepts on the pathogenesis of peri-implantitis: a narrative review. *Eur J Dent* 2021;**15**(2):379–87. https://doi.org/10.1055/s-0040-1721903
19. Rakic M, Galindo-Moreno P, Monje A, Radovanovic S, Wang HL, Cochran D, et al. How frequent does peri-implantitis occur? A systematic review and meta-analysis. *Clin Oral Investig* 2018;**22**(4):1805–16. https://doi.org/10.1007/s00784-017-2276-y
20. Albrektsson T, Canullo L, Cochran D, De Bruyn H. “Peri-implantitis”: a complication of a foreign body or a man-made “disease”. Facts and fiction. *Clin Implant Dent Relat Res* 2016;**18**(4):840–9. https://doi.org/10.1111/cid.12427
21. Wu X, Yan Q, Fang X, Hua F, Shi B, Tu YK. Spin in the abstracts of randomized controlled trials in periodontology and oral implantology: a cross-sectional analysis. *J Clin Periodontol* 2020:1079–86. https://doi.org/10.1111/jcpe.13340
22. Qin D, Hua F, Yue H, Yan Q, He H, Tu YK. The reporting and methodological quality of split-mouth trials in oral implantology: a methodological study. *Clin Oral Implants Res* 2022;**33**(12):1282–92. https://doi.org/10.1111/clr.14011
23. Cairo F, Sanz I, Matesanz P, Nieri M, Pagliaro U. Quality of reporting of randomized clinical trials in implant dentistry. A systematic review on critical aspects in design, outcome assessment and clinical relevance. *J Clin Periodontol* 2012;**39**(12):81–107. https://doi.org/10.1111/j.1600-051X.2011.01839.x
24. Buser D, Sennerby L, De Bruyn H. Modern implant dentistry based on osseointegration: 50 years of progress, current trends and open questions. *Periodontol 2000* 2017;**73**(1):7–21. https://doi.org/10.1111/prd.12185
25. Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ* 2021;**372**:n71. https://doi.org/10.1136/bmj.n71
26. Slim K, Nini E, Forestier D, Kwiatkowski F, Panis Y, Chipponi J. Methodological index for non-randomized studies (MINORS): development and validation of a new instrument. *ANZ J Surg* 2003;**73**(9):712–6. https://doi.org/10.1046/j.1445-2197.2003.02748.x
27. Atkins D, Best D, Briss PA, Eccles M, Falck-Ytter Y, Flottorp S, et al. Grading quality of evidence and strength of

recommendations. *BMJ* 2004; **328**(7454):1490. https://doi.org/10.1136/bmj.328.7454.1490
28. Ekelund JA, Lindquist LW, Carlsson GE, Jemt T. Implant treatment in the edentulous mandible: a prospective study on Brånemark system implants over more than 20 years. *Int J Prosthodont* 2003; **16**(6):602–8.
29. Attard NJ, Zarb GA. Long-term treatment outcomes in edentulous patients with implant-fixed prostheses: the Toronto study. *Int J Prosthodont* 2004; **17**(4):417–24.
30. Åstrand P, Ahlqvist J, Gunne J, Nilson H. Implant treatment of patients with edentulous jaws: a 20-year follow-up. *Clin Implant Dent Relat Res* 2008; **10**(4):207–17. https://doi.org/10.1111/j.1708-8208.2007.00081.x
31. Chappuis V, Buser R, Brägger U, Bornstein MM, Salvi GE, Buser D. Long-term outcomes of dental implants with a titanium plasma-sprayed surface: a 20-year prospective case series study in partially edentulous patients. *Clin Implant Dent Relat Res* 2013; **15**(6):780–90. https://doi.org/10.1111/cid.12056
32. Donati M, Ekestubbe A, Lindhe J, Wennström JL. Marginal bone loss at implants with different surface characteristics—a 20-year follow-up of a randomized controlled clinical trial. *Clin Oral Implants Res* 2018;**29**(5):480–7. https://doi.org/10.1111/clr.13145
33. Bakker MH, Vissink A, Meijer HJA, Raghoebar GM, Visser A. Mandibular implant-supported overdentures in (frail) elderly: a prospective study with 20-year follow-up. *Clin Implant Dent Relat Res* 2019;**21**(4):586–92. https://doi.org/10.1111/cid.12772
34. Thöne-Mühling M, Pricope L, Mogk M, Mengel R. Turned surface implants in patients treated for periodontitis: preliminary 10- to 20-year results of a long-term cohort study. *Int J Oral Implantol* 2020;**13**(2):173–84.
35. Jacobs R, Gu Y, Quirynen M, De Mars G, Dekeyser C, van Steenberghe D, et al. A 20-year split-mouth comparative study of two screw-shaped titanium implant systems. *Int J Oral Implantol* 2021; **14**(4):421–30.
36. Roccuzzo A, Imber JC, Marruganti C, Salvi GE, Ramieri G, Roccuzzo M. Clinical outcomes of dental implants in patients with and without history of periodontitis: a 20-year prospective study. *J Clin Periodontol* 2022;**49**(12):1346–56. https://doi.org/10.1111/jcpe.13716
37. Östman PO, Chrcanovic BR, Albrektsson T. A prospective report of the clinical outcome of TiUnite implants at 20 years of follow-up. *Int J Oral Maxillofac Implants* 2024(3):389–95. https://doi.org/10.11607/jomi.10654
38. Schünemann HJ, Cuello C, Akl EA, Mustafa RA, Meerpohl JJ, Thayer K, et al. GRADE guidelines: 18. How ROBINS-I and other tools to assess risk of bias in nonrandomized studies should be used to rate the certainty of a body of evidence. *J Clin Epidemiol* 2019; **111**:105–14. https://doi.org/10.1016/j.jclinepi.2018.01.012
39. Alshiddi IF. Survival rate and clinicoradiographic parameters around narrow-diameter dental implants for fixed dental prostheses in the posterior regions: a systematic review. *Dent Med Probl* 2023; **60**(2):345–53. https://doi.org/10.17219/dmp/140757
40. Jung RE, Zembic A, Pjetursson BE, Zwahlen M, Thoma DS. Systematic review of the survival rate and the incidence of biological, technical, and aesthetic complications of single crowns on implants reported in longitudinal studies with a mean follow-up of 5 years. *Clin Oral Implants Res* 2012;**23**(6):2–21. https://doi.org/10.1111/j.1600-0501.2012.02547.x
41. Albrektsson T, Donos N. Working Group 1. Implant survival and complications. The Third EAO Consensus Conference 2012. *Clin Oral Implants Res* 2012;**23**(6):63–5. https://doi.org/10.1111/j.1600-0501.2012.02557.x
42. Howe MS, Keys W, Richards D. Long-term (10-year) dental implant survival: a systematic review and sensitivity meta-analysis. *J Dent* 2019;**84**:9–21. https://doi.org/10.1016/j.jdent.2019.03.008
43. Lupi SM, Albini B, Baena ARY, Lanfrè G, Galinetto P. Anatase forming treatment without surface morphological alteration of dental implant. *Materials* 2020;**13**(22):1–14. https://doi.org/10.3390/ma13225280
44. Lupi SM, Galinetto P, Albini B, Ronza ED, Rizzo S, Baena RR. Micro-Raman spectroscopy of dental implants subjected to different surface treatments. *Appl Sci* 2020;**10**(7):2417. https://doi.org/10.3390/app10072417
45. Matsuura T, Komatsu K, Cheng J, Park G, Ogawa T. Beyond microroughness: novel approaches to navigate osteoblast activity on implant surfaces. *Int J Implant Dent* 2024;**10**(1):35. https://doi.org/10.1186/s40729-024-00554-x
46. Herrmann H, Kern JS, Kern T, Lautensack J, Conrads G, Wolfart S. Early and mature biofilm on four different dental implant materials: an in vivo human study. *Clin Oral Implants Res* 2020;**31**(11):1094–104. https://doi.org/10.1111/clr.13656
47. Albrektsson T, Zarb G, Worthington P, Eriksson AR. The long-term efficacy of currently used dental implants: a review and proposed criteria of success. *Int J Oral Maxillofac Implants* 1986; **1**(1):11–25.
48. Kupka JR, König J, Al-Nawas B, Sagheb K, Schiegnitz E. How far can we go? A 20-year meta-analysis of dental implant survival rates. *Clin Oral Investig* 2024; **28**(10):541. https://doi.org/10.1007/s00784-024-05929-3
49. Schimmel M, Srinivasan M, McKenna G, Müller F. Effect of advanced age and/or systemic medical conditions on dental implant survival: a systematic review and meta-analysis. *Clin Oral Implants Res* 2018;**29**(16):311–30. https://doi.org/10.1111/clr.13288

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